Mispricing and Arbitrage in Prediction Markets

Probability coherence, order-book liquidity and execution costs on Kalshi

Joel Cerraga

Quantitative research project

October 2026

Final assembled paper

Abstract

Prediction-market prices invite a probability interpretation, but a discrepancy between prices is not sufficient evidence of an executable arbitrage. Contract definitions, quoted liquidity, fees, observation timing and incomplete execution determine which apparent opportunities survive. This study develops a read-only research engine for Kalshi that combines explicit payoff-state checks, fixed-point order-book reconstruction, quantity-dependent acquisition costs and chronological probability evaluation.

The historical sample comprises 270 daily maximum-temperature events and 1,620 contracts across New York, Chicago and Miami between 19 July and 16 October 2026. Public REST collection provides 56,006 hourly candles. The primary noon specification retains 268 complete six-contract event panels: 162 for development, 53 for validation and 53 for later evaluation. A separate live pilot records 720 individual books across 36 contracts in 20 batch responses over approximately three minutes. Synthetic experiments test known payoff inconsistencies and the consequences of incomplete multi-leg execution.

There are 34 positive gross historical basket-side gaps among 536 conditional checks, but none remains positive under the stated current-fee scenario. The historical records do not establish executable depth or fills. In the live pilot, 378 basket checks have sufficient displayed depth and no positive net gap; 342 fail the depth requirement. Forty-six complete two-sided event observations support statewise optimisation, with no positive rechecked candidate. A conservative full-precision payoff audit and exceptional settlement provisions further limit unconditional arbitrage certification.

A one-parameter probability calibration fits a development temperature of 1.0344 but performs worse than the unchanged market-derived baseline on validation. The selection rule therefore retains the baseline, whose later noon log loss is 0.718589 nats and multiclass Brier score is 0.408273. Synthetic execution stresses show that positive expected value can coexist with losses from incomplete baskets. The contribution is a reproducible framework for distinguishing probability disagreement, conditional payoff gaps and executable opportunities, rather than evidence of realised trading profitability. The discussion records acquisition failures, contract-audit corrections, model rejection and remaining data-access and execution limitations.

Keywords: prediction markets; market microstructure; probability calibration; arbitrage screening; order-book liquidity; execution risk.

Table of contents

Abstract i

List of figures iv

List of tables iv

List of equations v

List of code listings v

List of abbreviations vi

List of mathematical symbols vii

1 Introduction and research context 1

1.1 Motivation . . . 1

1.2 Aim and research questions . . . 1

1.3 Contribution and scope . . . 1

2 Research foundations and interpretation 2

2.1 Prices, probabilities and market frictions . . . 2

2.2 Depth and order-flow evidence . . . 2

2.3 Probability evaluation and execution . . . 3

3 Public data and chronological design 3

3.1 Market universe and source roles . . . 3

3.2 Completed-hour observation rule . . . 4

3.3 Development, validation and later samples . . . 4

3.4 Provenance and reproducibility . . . 5

4 Contract payoffs and order-book microstructure 6

4.1 Binary claims and settlement states . . . 6

4.2 Complementary bids and asks . . . 7

4.3 Displayed imbalance and weighted prices . . . 7

4.4 Quantity-dependent acquisition costs . . . 8

4.5 Fees, rounding and fill granularity . . . 8

5 Payoff constraints and arbitrage screening 9

5.1 Conditional basket relationships . . . 9

5.2 Statewise net payoff . . . 9

5.3 Relaxation, rechecking and reporting gates . . . 10

6 Probability modelling and validation 11

6.1 Market-derived probability baseline . . . 11

6.2 Temperature calibration and selection . . . 11

6.3 Proper scores and expected-value diagnostics . . . 12

6.4 Paired date-block uncertainty . . . 12

7 Empirical findings 13

7.1 Historical spreads and conditional basket gaps . . . 13

7.2 Calibration fit and model rejection . . . 14

7.3 Reliability and uncertainty . . . 15

7.4 Observation-time and city sensitivity . . . 16
7.5 Live liquidity and statewise screening . . . 18
7.6 Exploratory imbalance diagnostic . . . 19
8 Synthetic validation and execution risk 20
8.1 Known-truth screening mechanisms . . . 20
8.2 Incomplete multi-leg execution . . . 20
9 Discussion, research reflections and limitations 22
9.1 What the combined evidence establishes . . . 22
9.2 Acquisition failures and implemented corrections . . . 23
9.3 Contract and model assumptions that failed . . . 23
9.4 Verification and residual risk . . . 23
9.5 Data access and publication status . . . 24
9.6 A stronger subsequent research design . . . 24
10 Conclusion 25
References 26
A Reproducibility and implementation 27
A.1 Notebook sequence and recorded inputs . . . 27
A.2 Fee calculation excerpt . . . 28
A.3 Statewise optimisation excerpt . . . 28
A.4 Chronological calibration excerpt . . . 29
B Interactive research explorers 30
B.1 Recorded liquidity in three dimensions . . . 30
B.2 Calibration sensitivity . . . 30
B.3 Synthetic basket execution . . . 31

List of figures

1 Completed-hour sample coverage . . . 5
2 Historical quoted spreads . . . 13
3 Gross and fee-adjusted basket gaps . . . 14
4 Later-period probability scores . . . 15
5 Descriptive probability reliability . . . 16
6 Date-block interval sensitivity . . . 17
7 Quantity-dependent acquisition costs . . . 19
8 Snapshot imbalance and subsequent quotes . . . 19
9 Synthetic screening mechanisms . . . 21
10 Synthetic partial-execution stress . . . 22
11 Recorded order-book depth in three dimensions . . . 30
12 Probability calibration sensitivity surface . . . 31
13 Analytical synthetic loss-probability surface . . . 31

List of tables

1 Abbreviations . . . vi
2 Mathematical symbols . . . vii
3 Empirical market universe . . . 3
4 Acquisition routes and analytical roles . . . 4
5 Chronological samples and completed-hour coverage . . . 5
6 Payoff partition and settlement audit . . . 6
7 Illustrative fee arithmetic . . . 9
8 Historical conditional basket discrepancies . . . 13
9 Primary probability scores at noon . . . 14
10 Paired date-block sensitivity for the rejected candidate . . . 16
11 Later-period sensitivity to observation time . . . 17
12 Later-period scores by city . . . 17
13 Recorded live-pilot audit . . . 18
14 Known-truth synthetic screening checks . . . 20
15 Synthetic six-leg execution stress . . . 21
16 Research and presentation verification . . . 24
17 Executed notebook sequence . . . 27

List of equations

1 Binary payoff 6
2 Complementary order-book quotes 7
3 Quoted spread and midpoint 7
4 Displayed depth imbalance 7
5 Depth-weighted midpoint 7
6 Depth-aware acquisition cost 8
7 Quadratic taker fee before rounding 8
8 Fee and balance rounding 8
9 Conditional probability coherence 9
10 Statewise net payoff 10
11 Finite-state portfolio programme 10
12 Market-derived baseline 11
13 Probability temperature calibration 11
14 Multiclass logarithmic loss 12
15 Multiclass Brier score 12
16 Model-based expected edge 12
17 Date-paired score difference 12
18 Synthetic complete-basket fill probability 21

List of code listings

1 Fill rounding and rebate accumulator 28
2 Statewise portfolio screening 29
3 Development-only temperature fitting 29

List of abbreviations

Abbreviation	Meaning
API	Application programming interface
CI	Confidence interval
CSV	Comma-separated values
FCM	Futures commission merchant
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LP	Linear programming
NWS	National Weather Service
NYC	New York City
OHLC	Open, high, low and close
PDF	Portable Document Format
PnL	Profit and loss
REST	Representational state transfer
SHA-256	Secure Hash Algorithm, 256-bit
TWC	The Weather Company
USD	United States dollar
UTC	Coordinated Universal Time
VWAP	Volume-weighted average price

Table 1. *Abbreviations.*

List of mathematical symbols

Symbol	Meaning	Units
i, j	Contract or outcome index	index
e	Daily city event	index
d	Calendar date	index
ℓ	Price level or enumerated fill	index
ω	Contract resolution state	state
Ω	Set of included resolution states	set
A_i	Event specified by contract i	set
X_i, X_i^N	YES and NO terminal payoffs	USD / contract
$\mathbf{1}\{\cdot\}$	Indicator of a true condition	0 or 1
a_i^Y, a_i^N	Best YES and NO asks	USD / contract
b_i^Y, b_i^N	Best YES and NO bids	USD / contract
s_i, m_i	Quoted spread and midpoint	USD / contract
$\widetilde{m}_i$	Depth-weighted midpoint	USD / contract
D_i^b, D_i^a	Displayed quantity at the best bid and ask	contracts
I_i	Snapshot depth imbalance	unitless
$q_i, q_{i\ell}$	Total acquired quantity and quantity per fill	contracts
$p_{i\ell}$	Acquisition price at level or fill ell	USD / contract
$C_i(q_i)$	Purchase cost excluding fees	USD
f_i^{raw}	Unrounded model fee for an order	USD
$F_i(q_i)$	Total fee after rounding and rebates	USD
M_i	Series fee multiplier	unitless
u_ℓ, t_ℓ	Signed fill revenue and rounded trade fee	USD
δ	Target balance precision	USD
r_ℓ, R_ℓ	Balance rounding and accumulator rebate	USD
H_ℓ	Carried rounding accumulator	USD
$\lceil x \rceil_h, \lfloor x \rfloor_h$	Round x up or down to a multiple of h	same as x
$\Pi(\omega; \mathbf{q})$	Portfolio net payoff in a state	USD
$\mathbf{A}$	State-by-position payoff matrix	USD / contract
$\mathbf{c}, \mathbf{d}$	Unit acquisition costs and displayed depth caps	USD / contract; contracts
B, g	Budget and lowest statewise net payoff	USD
$p_{ei}, \hat{p}_{ei}$	Baseline and model probability for an outcome	probability
ϵ	Positive midpoint floor (0.0001)	USD / contract
T	Probability calibration temperature	unitless
y_e	Observed winning outcome for event e	index
N, D, n_d	Events, dates and accepted events on date d	counts
$L_{\log}, L_B$	Mean logarithmic loss and Brier score	nats; unitless
$\widehat{V}_i(q)$	Model-based expected net payoff	USD
$\overline{\Delta}$	Equal-date paired score difference	score units
K, ρ	Basket legs and per-leg fill probability	count; probability
J_i	Synthetic indicator that leg i fills	0 or 1
η	Additional entry cost per filled synthetic leg	USD / contract

Table 2. *Mathematical symbols.*

1 Introduction and research context

1.1 Motivation

A binary prediction-market contract links a financial payoff to an observable event. This makes the market an appealing setting for probability modelling: a quoted price resembles a probability, and related contracts create arithmetic relationships that appear easy to test. The apparent simplicity can be misleading. A midpoint may be unavailable to trade, a displayed quantity may disappear, fees may exceed a small discrepancy, and apparently complementary market titles may conceal different settlement conditions.

As Wolfers and Zitzewitz (2006) explain in *Interpreting prediction market prices as probabilities*, the relationship between a market price and a probability depends on assumptions about beliefs and preferences. A price is therefore treated here as a market-derived reference rather than an observed physical probability. As Hasbrouck (2007) develops in *Empirical market microstructure*, trading institutions and the process through which orders meet are central to the interpretation of observed prices. These ideas motivate an analysis that connects probability estimates to the actual cost and feasibility of acquiring positions.

The project focuses on daily maximum-temperature markets in three US cities. Their recurring dates and ordered contract ranges make event membership, chronological evaluation and payoff relationships inspectable. The study does not generalise its settlement adapter to elections, sporting events or unrelated economic contracts. Those applications would require separate rule interpretation and outcome modelling.

1.2 Aim and research questions

The aim is to build and assess a reproducible engine that separates statistical disagreement from structural payoff inconsistencies and execution feasibility. Three questions organise the investigation. First, do related contracts exhibit price relationships that imply a positive lowest-state payoff under a stated contract model? Second, do such relationships survive fees, available depth and observation uncertainty? Third, can a calibrated probability model improve on a market-derived baseline when transferred to later dates?

These questions require different evidence. Probability accuracy is evaluated against subsequent outcomes. Structural screening evaluates every state in a supplied payoff matrix. Execution feasibility concerns prices, quantities and fills at a common trading opportunity. An improvement in one layer cannot be substituted for evidence at another. In particular, a positive model-based expected value can occur alongside a negative payoff in some states.

1.3 Contribution and scope

The resulting Python engine collects public REST responses, reconstructs complementary quotes, calculates acquisition costs with decimal arithmetic and checks conditional baskets and general long-only payoff portfolios. It also fits a deliberately simple calibration model under a fixed chronological selection rule. Eight executed notebooks connect

the calculations to data provenance, synthetic checks and research reflections. Three interactive explorers provide views of displayed depth, calibration sensitivity and synthetic execution risk.

The empirical contribution is bounded. The historical analysis uses hourly quote summaries, while the repeated books come from one short session. No orders are submitted, no account credentials are required by the collector, and no fill history is available. The work consequently evaluates research logic and observed quote conditions rather than a deployed trading strategy. The paper preserves null and rejected results because they show where apparent opportunities fail the available evidence.

2 Research foundations and interpretation

2.1 Prices, probabilities and market frictions

The probability interpretation is useful for organising forecasts, but it is not an assertion that every quoted midpoint is a calibrated belief. Risk preferences, heterogeneous information, trading constraints and the spread can all affect the price at which a position is available. The baseline in this study normalises the positive midpoints within a complete event. That transformation creates a probability vector for scoring; it does not make all transformed components executable prices.

As Glosten and Milgrom (1985) show in *Bid, ask and transaction prices in a specialist market with heterogeneously informed traders*, asymmetric information can generate a bid–ask spread even under competitive liquidity provision. Their theoretical setting does not establish the cause of every spread in the present sample. It does explain why a midpoint comparison omits an economically meaningful trading friction. A buyer crosses to the ask and must additionally account for fees and the available size at that price.

The research therefore uses the word *mispricing* in two qualified senses. A model disagreement is a difference between an estimated probability and a cost of acquisition. A structural discrepancy is a relationship between several quotes that fails a stated payoff identity. Neither label, by itself, implies realised profit or a risk-free trading instruction.

2.2 Depth and order-flow evidence

As Cont, Kukanov and Stoikov (2014) report in *The price impact of order book events*, order-book changes and available depth help explain short-horizon price movements. Their order-flow imbalance is constructed from order events. The present REST observations contain snapshots, so the implemented imbalance measures displayed quantities at the best bid and ask. The two measurements are not interchangeable.

This distinction matters when interpreting the empirical scatter plot. Between two snapshots, orders can be entered, cancelled, modified or executed without appearing separately in the retained data. A correlation between snapshot imbalance and the next observed midpoint is therefore descriptive. It cannot identify a causal price-impact coefficient, reconstruct queue priority or establish a feasible high-frequency predictor.

2.3 Probability evaluation and execution

As Gneiting and Raftery (2007) explain in *Strictly proper scoring rules, prediction, and estimation*, probability forecasts can be evaluated as full distributions. The study uses logarithmic loss and the multiclass Brier score, with the event as the scoring unit. A model is selected on validation loss before its later scores are calculated. This permits an honest rejection of an unnecessary calibration step.

As Avellaneda and Stoikov (2008) discuss in *High-frequency trading in a limit order book*, execution and inventory exposure are integral to trading decisions. The present work does not estimate their market-making model. It uses a simpler synthetic experiment to demonstrate how separately filled legs can leave a residual directional position even when a fully acquired basket has a positive terminal margin.

3 Public data and chronological design

3.1 Market universe and source roles

Table 3 defines the empirical universe. The series were chosen for repeated daily events and ordered temperature ranges before model fitting. Each historical event contains six contracts. A contract is the individual range or tail claim, while an event is the city-date outcome shared by those contracts. These units must remain distinct when counting observations.

City	Series	Historical dates	Events / contracts
New York	KXHIGHNY	19 July–16 October 2026	90 / 540
Chicago	KXHIGHCHI	19 July–16 October 2026	90 / 540
Miami	KXHIGHMIA	19 July–16 October 2026	90 / 540

Table 3. *Empirical market universe.*
Daily maximum-temperature events. Six contracts per event; 270 historical events and 1,620 contracts in total.

As the Kalshi (2026a) API documentation describes, selected market-data routes are publicly readable without authentication. Table 4 records the roles of the routes used here. The collection code issues GET requests and retains request timing, response metadata and parsed bodies. It contains no order-submission method. Technical access and permission to retain or publish the returned data are separate matters, discussed in Section 9.5.

Current and archived market records are combined and deduplicated by ticker before selecting the 90-date window. This avoids interpreting the current endpoint's retention boundary as the beginning of the market's history. A live batch-candle route supplies available hourly records, and an archived per-market route supplies otherwise missing history. The collected total is 56,006 hourly candles across the 1,620 selected contracts; it is not an uninterrupted exchange-wide order-event history.

Material	Public REST route	Analytical role
Market metadata	`/markets`; `/historical/markets`	Event membership, strikes, settled outcomes and rule text.
Hourly candles	`/markets/candlesticks`; historical market candles	Completed-hour bid and ask summaries; no historical depth.
Live books	`/markets/orderbooks`	Repeated fixed-point bids and quantities; request timing recorded.
Fee metadata	`/series/{series}`; fee-change records	Series fee type and multiplier; dated historical applicability remains uncertified.

Table 4. *Acquisition routes and analytical roles.*
Base endpoint: https://external-api.kalshi.com/trade-api/v2. All research collection uses GET requests.

3.2 Completed-hour observation rule

The primary observation is the hourly candle ending exactly at 12:00 in the event city's local time. New York and Miami use the America/New_York time zone, while Chicago uses America/Chicago. The latest eligible candle is first restricted to an endpoint no later than the cutoff; an exact completed-hour endpoint is then required. A stale earlier candle is not carried forward. The 09:00 and 15:00 observations use the same rule as descriptive sensitivity checks.

Every accepted event requires valid bid and ask values for all six contracts, with prices between zero and one and the bid no greater than the ask. There is no interpolation or substitution of a later observation. For probability scoring, the event must subsequently have exactly one binary winning contract. Settled outcomes are kept separately from feature construction; they determine the score, not the forecast inputs. Final market volume, open interest and settlement price do not enter the probability vector.

This construction is retrospective. Historical candles are acquired after their event dates, and the complete labelled dataset exists locally during processing. The separation of features, model fitting and later scoring is enforced by code and recorded hashes, not by physical inability to access the outcomes. The evaluation is consequently described as chronological held-out research rather than a prospective or externally preregistered trading trial.

3.3 Development, validation and later samples

All cities on a calendar date remain in the same chronological partition. Development covers 19 July–10 September 2026, validation covers 11–28 September, and the later test period covers 29 September–16 October. Splitting individual contracts at random would leak closely related event information across samples and is avoided.

The noon specification retains 268 of 270 planned events. The two exclusions are Chicago on 12 September and 5 October, both because an exact completed-hour observation is missing. This leaves 162 development, 53 validation and 53 later events. The later events span 18 dates. Figure 1 locates the missing panels rather than concealing them inside an aggregate count.

Sample	2026 dates	Planned	09:00	12:00	15:00
Development	19 July–10 September	162	159	162	159
Validation	11–28 September	54	54	53	53
Test	29 September–16 October	54	54	53	53
Total	90 dates	270	267	268	265

Table 5. *Chronological samples and completed-hour coverage.*
Counts are complete six-contract event panels at city-local time. The noon specification is primary.

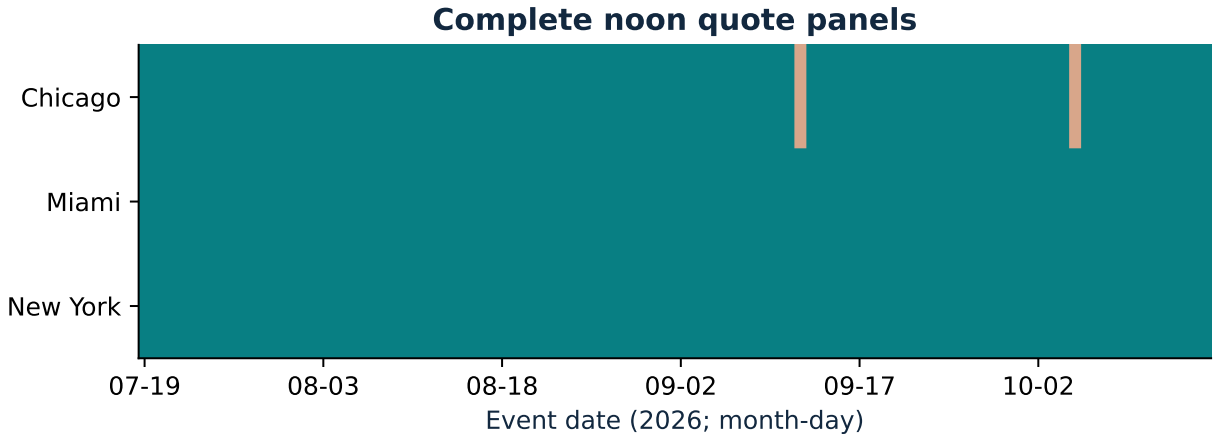


Figure 1. *Coverage of complete six-contract panels at 12:00 local time; peach indicates an excluded event.*

At 09:00, 267 event panels are accepted; at 15:00, 265 are accepted. These are overlapping observations of the same event universe, not independent replications. Differences across hours combine a changing information horizon with modest changes in sample membership. The primary hour and chronological split remain fixed after these differences are observed.

3.4 Provenance and reproducibility

The source cache contains 521 stored snapshots. Each retains the URL, request start and response-received times, response information, the parsed JSON body and an observed hash of the original response bytes. A regenerated index adds hashes of the stored snapshot file and a canonical JSON representation of the body. The original wire bytes are not stored verbatim, so their recorded hash cannot be independently recomputed from the parsed object alone. The file and canonical-payload hashes serve separate reproducibility purposes.

An offline reproduction verifies the saved inputs and model freeze before rebuilding outputs. Model fitting uses only the development outcomes, selection uses validation loss, and the final prediction file is hashed before later scoring. The freeze is an internal chronology record. It guards against silent replacement of the retained model and predictions; it does not transform retrospective analysis into prospective evidence.

4 Contract payoffs and order-book microstructure

4.1 Binary claims and settlement states

For ordinary binary resolution, let A_i be the precisely defined event underlying contract i, and let ω be a resolution state. YES and NO payoffs, measured in dollars per unit-payout contract, satisfy

$$X_i(\omega) = \mathbf{1}\{\omega \in A_i\}, \qquad X_i^N(\omega) = 1 - X_i(\omega) \tag{1}$$

Equation 1. Binary payoff

The abbreviated market title is insufficient to determine A_i. Location, date, source agency, range endpoints and exceptional provisions must all agree before relationships across contracts can be asserted. Equation 1 describes ordinary binary outcomes; an exceptional scalar or fair-price settlement requires the actual payout rule to be represented separately.

As Kalshi (2026e) specifies in the GLOBALTEMPERATURE terms, settlement follows the source's reported precision, and missing data can invoke a fair-price contingency. The engine therefore audits both integer-temperature partitions and a conservative set of full-precision boundary and between-boundary states. This audit does not infer that a noninteger gap settlement occurred. It tests whether a guarantee can be justified without an additional proof about the admissible source values.

Check	Historical	Live	Interpretation
Daily events	270	6	No mixing of city/date event membership.
Integer partitions	270	6	Exactly one winning YES in the model restricted to integers.
Exhaustive over full-precision probes	0	0	A conservative continuous-state audit retains an all-zero YES state.
Observed single binary winner	270	Not settled	Historical scoring is conditional on this observed outcome universe.

Table 6. *Payoff partition and settlement audit.*

Continuous-state incompleteness is a screening assumption, not evidence that a noninteger gap resolution actually occurred.

For all 276 historical and live events, the six ranges partition the model restricted to integer temperatures. When the declared comparison operators are evaluated at every finite endpoint, between successive endpoints and beyond the tails, the audit also finds an all-zero YES state. The resulting seven distinct rows consist of the six single-winner states and that additional row. If the applicable official source is proven to report only admissible integer values, some continuous probes could be unreachable. That narrower admissible domain has not been certified for every retained contract.

The conservative state set may therefore reject opportunities that a more fully documented contract model would allow. This is a deliberate limit on certification, not a finding that the exchange's contracts are defective. Exceptional settlement remains a further contingency

outside the ordinary temperature matrix. Historical scoring conditions on the observed single-winner universe and does not claim to model the probability of every exceptional outcome.

4.2 Complementary bids and asks

The order-book response supplies YES bids and NO bids as fixed-point dollar and quantity strings. Following the complementary relationship documented by Kalshi (2026a), the opposite-side bids imply the asks:

$$a_i^Y = 1 - b_i^N, \qquad a_i^N = 1 - b_i^Y \tag{2}$$

Equation 2. Complementary order-book quotes

Prices and quantities are parsed with Python's Decimal type. Repeated price levels are aggregated and prices are sorted explicitly. The implementation rejects non-finite values, negative quantities and invalid sides. Empty arrays are represented as unavailable liquidity, not as a zero-price trading opportunity.

When both sides are available and uncrossed, the quoted spread and midpoint are

$$s_i = a_i^Y - b_i^Y, \qquad m_i = (a_i^Y + b_i^Y)/2 \tag{3}$$

Equation 3. Quoted spread and midpoint

The midpoint is a descriptive reference. It is not the price charged by the acquisition-cost routine. A missing bid or ask prevents the calculation of a two-sided spread, even when one side can still supply contracts for a particular purchase.

4.3 Displayed imbalance and weighted prices

Let D_i^b and D_i^a denote quantities at the best YES bid and implied YES ask. The displayed imbalance is

$$I_i = (D_i^b - D_i^a)/(D_i^b + D_i^a) \tag{4}$$

Equation 4. Displayed depth imbalance

It lies between minus one and one when the denominator is positive. A positive value indicates more displayed quantity at the best bid than at the best ask. The calculation omits deeper levels, hidden liquidity and all events between snapshots.

A descriptive depth-weighted midpoint places more weight on the ask when bid depth is larger:

$$\widetilde{m}_i = (a_i^Y D_i^b + b_i^Y D_i^a)/(D_i^b + D_i^a) \tag{5}$$

Equation 5. Depth-weighted midpoint

Equal depths recover the ordinary midpoint. This expression is an algebraic summary, not an estimated efficient-price model. No claim that it predicts a later trade price is inferred from its definition. The empirical analysis retains the subsequent midpoint comparison as an exploratory diagnostic only.

4.4 Quantity-dependent acquisition costs

To acquire q contracts, the engine consumes ascending ask levels until the requested quantity is reached. With $q_{i\ell}$ units taken at price $p_{i\ell}$, the purchase cost is

$$C_i(q) = \sum_{\ell} p_{i\ell} q_{i\ell}, \qquad \sum_{\ell} q_{i\ell} = q \tag{6}$$

Equation 6. Depth-aware acquisition cost

Each fill quantity is bounded by the displayed depth at its level. If total depth is insufficient, the requested sweep is rejected and its shortfall recorded; the cost of a partial fill is not presented as the cost of a complete basket. Dividing $C_i(q)$ by q gives the volume-weighted acquisition price. Its difference from the best ask measures displayed-depth slippage for that hypothetical sweep.

The historical candles have no accompanying depth. They can support one-unit quote arithmetic but cannot run the same executable-size test. Even the live sweep is a static-book calculation: it does not establish that all displayed orders remain available while an incoming order is processed.

4.5 Fees, rounding and fill granularity

The current quadratic taker schedule and series multiplier are obtained from Kalshi (2026d) and the retained series metadata. Before rounding, the order's fee is

$$f_i^{\mathsf{raw}} = 0.07 M_i \sum_{\ell} q_{i\ell} p_{i\ell} (1 - p_{i\ell}). \tag{7}$$

Equation 7. Quadratic taker fee before rounding

The audited series have a multiplier of one. The engine rejects unsupported fee types and missing multipliers rather than silently applying a generic fee. It models acquisition as a taker operation; rebates associated with a separate market-making strategy are not assumed.

The Kalshi (2026c) fee-rounding documentation distinguishes the trade-fee precision from the balance grid. The recurrence below states the implemented order-level calculation, with $H_0 = 0$:

$$\begin{aligned} t_\ell &= \lceil f_\ell^{\mathsf{model}} \rceil_{10^{-6}}, \quad u_\ell = -p_\ell q_\ell, \\ r_\ell &= (u_\ell - t_\ell) - \lfloor u_\ell - t_\ell \rfloor_\delta, \\ R_\ell &= \mathsf{min}\{\lfloor H_{\ell-1} + r_\ell \rfloor_\delta, \lfloor t_\ell + r_\ell \rfloor_\delta\}, \\ H_\ell &= H_{\ell-1} + r_\ell - R_\ell, \quad f_\ell^{\mathsf{net}} = t_\ell + r_\ell - R_\ell. \end{aligned} \tag{8}$$

Equation 8. Fee and balance rounding

Here f_ℓ^{model} is the raw fee of fill ℓ, u_ℓ is its signed purchase revenue, and $\lfloor \cdot \rfloor_\delta$ rounds down to the balance grid. Direct-member and non-direct-member scenarios use $\delta = 0.0001$ and $\delta = 0.01$ dollars respectively. The carried accumulator prevents treating each balance adjustment as unrelated to earlier fills of the same order. The rounded order fee $F_i(q_i)$ is the sum of the fill net fees.

Scenario	Purchase cost	Fee	Total outlay
One contract at \$0.50; direct precision	\$0.5000	\$0.0175	\$0.5175
One contract at \$0.50; cent precision	\$0.5000	\$0.0200	\$0.5200
Six one-contract legs at \$0.15; raw fee	\$0.90000	\$0.05355	\$0.95355

Table 7. *Illustrative fee arithmetic.*

Constructed examples with fee multiplier one. The last row is the raw-fee synthetic experiment, without fill-level balance rounding.

A displayed price level can aggregate several counterparties and therefore need not correspond to one execution fill. The default sweep treats a consumed level as a fill for the fee scenario; the engine can fragment it into smaller assumed fills. Neither choice reconstructs undisclosed counterparty matching. The historical fee calculation is also a current-schedule scenario, because the complete applicable fee and balance regime has not been certified for every historical date.

5 Payoff constraints and arbitrage screening

5.1 Conditional basket relationships

For a verified mutually exclusive, exhaustive collection of unit-payout outcomes, a frictionless probability vector can lie within the quoted intervals only if

$$\sum_i b_i^Y \leq 1 \leq \sum_i a_i^Y \tag{9}$$

Equation 9. Conditional probability coherence

The condition is a useful quote-coherence check. It depends on a valid partition and does not assert that a trader can transact at every quote. If all YES asks sum to less than one, an equal-quantity YES basket has a positive gross ordinary-resolution margin. If all YES bids sum to more than one, the complementary equal-quantity NO basket is the analogous long-only construction. With K ordinary single-winner outcomes, one unit of every NO claim pays $K - 1$ dollars.

The engine evaluates both sides and then subtracts the quantity-dependent costs and fees. For the temperature study, these equal-basket summaries are explicitly conditional on integer, single-winner resolution. A non-exhaustive state set or exceptional payout cannot be repaired by merely renormalising the prices.

5.2 Statewise net payoff

Let **q** contain long positions and let each position's payoff be represented across the included states. The portfolio's net payoff is

$$\Pi(\omega;\mathbf{q}) = \sum_i X_i(\omega) q_i - \sum_i [C_i(q_i) + F_i(q_i)] \tag{10}$$

Equation 10. Statewise net payoff

A strictly positive minimum of $\Pi(\omega;\mathbf{q})$ over the supplied states is a structural candidate within that model. It is not an unconditional certificate if states are missing, and it does not establish that the quantities were filled. Conversely, a model probability can assign a positive expectation to a portfolio whose minimum payoff is negative. That portfolio is a risky value trade rather than a statewise arbitrage.

As Boyd and Vandenberghe (2004) explain in *Convex optimization*, linear constraints can express a feasible portfolio under resource and payoff bounds. The implemented screen uses a state-by-position matrix **A**, fee-adjusted unit costs **c**, available-depth bounds **d** and budget B:

$$\begin{aligned} &\max_{\mathbf{q},g} \; g \\ &\text{subject to } \; \mathbf{Aq} - (\mathbf{c}^{\mathsf{T}}\mathbf{q})\mathbf{1} \geq g\mathbf{1}, \\ &\qquad\qquad 0 \leq \mathbf{q} \leq \mathbf{d}, \quad \mathbf{c}^{\mathsf{T}}\mathbf{q} \leq B, \quad g \geq 0. \end{aligned} \tag{11}$$

Equation 11. Finite-state portfolio programme

The matrix includes YES and NO positions for each contract. Their rows are built from the same underlying temperature state rather than treating each claim's outcome as independent. The zero portfolio is feasible, so zero is a legitimate optimum when no positive lowest-state payoff can be supported. A solver returning zero is not an optimisation failure.

5.3 Relaxation, rechecking and reporting gates

The live implementation uses top-of-book unit prices and raw fee slopes in Equation 11, with a budget of 100 dollars. This is a continuous screening relaxation. It bounds each position by displayed top-level depth, floors the resulting quantities to the 0.01-contract grid, then recomputes purchase costs, rounded fees and all state payoffs. A positive relaxed result would not be reported as surviving unless the rechecked portfolio also remained positive.

The live sample additionally lacks exchange timestamps for individual books. A two-second request-span threshold is retained as a strict observation screen, but a fast response would still not prove atomic cross-contract prices. A batch response can collect several books without guaranteeing that every book reflects the same exchange instant. Settlement uncertainty and the absence of actual simultaneous fills remain separate reasons to withhold an executable-arbitrage label.

This separation also prevents circular use of YES and NO positions. Complementary exposures must be priced through the observed opposite-side liquidity and represented in one state matrix. They are not treated as independent risk-free resources. The general optimiser accepts a supplied payoff matrix; the empirical rule adapter remains limited to the three audited temperature series.

6 Probability modelling and validation

6.1 Market-derived probability baseline

For each accepted event, positive midpoints are normalised using a floor $\epsilon = 0.0001$:

$$p_i = \max(m_i, \epsilon) / \sum_j \max(m_j, \epsilon) \tag{12}$$

Equation 12. Market-derived baseline

The floor avoids a zero probability and an undefined logarithmic score. It is applied before normalisation and is held fixed across samples. The baseline conditions on the observed six-outcome single-winner event universe. It is not a forecast for missing-data contingencies or scalar settlements.

Normalisation removes the midpoint sum's deviation from one for scoring purposes. It does not remove that deviation from the separate quote-coherence audit. Keeping the two calculations separate is essential: forcing a statistical vector onto the simplex should not erase an observed spread, change a purchase price or establish a contractual partition.

6.2 Temperature calibration and selection

As Guo et al. (2017) describe in *On calibration of modern neural networks*, a temperature parameter can alter the concentration of a probability vector. The project applies that transformation to the logarithms of the market-derived baseline:

$$\hat{p}_i(T) = \frac{\exp(\log p_i / T)}{\sum_j \exp(\log p_j / T)} \tag{13}$$

Equation 13. Probability temperature calibration

At $T = 1$, the transformation is the identity. A temperature above one makes the distribution less concentrated, while a temperature below one sharpens it. Outcome ordering is unchanged. No neural network is fitted, and the neural-network performance findings in Guo et al. are not assumed to transfer to prediction-market prices.

The development fit minimises mean noon logarithmic loss over $0.25 \leq T \leq 4$, using a bounded optimisation in log temperature. The fitted candidate is then compared with the unchanged baseline on validation loss. The baseline wins ties, including the implementation's numerical tolerance. Only the winning method is designated as selected. A later comparison of the rejected candidate is supplementary and cannot revise the recorded selection.

6.3 Proper scores and expected-value diagnostics

For N events with observed winner y_e, the logarithmic loss is

$$L_{\log} = -\frac{1}{N}\sum_{e}\log \hat{p}_{e,y_e} \tag{14}$$

Equation 14. Multiclass logarithmic loss

Natural logarithms give nats per event, and lower loss is better. A forecast that assigns very low probability to the realised winner receives a large penalty. As Brier (1950) proposed in *Verification of forecasts expressed in terms of probability*, squared probability errors provide another forecast-verification criterion. The present multiclass form sums over the outcomes within each event:

$$L_B = \frac{1}{N}\sum_{e}\sum_{i}(\hat{p}_{ei} - \mathbf{1}\{y_e = i\})^2 \tag{15}$$

Equation 15. Multiclass Brier score

This summed Brier convention has a range from zero to two for a categorical outcome. It is not divided by the number of contracts. The event remains the unit of observation; the six outcomes are jointly constrained and cannot be counted as six independent forecasts.

To connect a probability estimate to a trading hypothesis, the expected net payoff of buying a YES claim is

$$\widehat{V}_i(q) = q\hat{p}_i - C_i(q) - F_i(q) \tag{16}$$

Equation 16. Model-based expected edge

For a NO claim, $\hat{p}_i$ is replaced by $1 - \hat{p}_i$. Positive expected payoff still depends on the estimated probability and permits losses in individual states. The exploratory historical value screen chooses at most one claim per later noon event when estimated one-contract edge exceeds two cents under the raw current-fee scenario. It is a diagnostic using candle asks, not an execution backtest.

6.4 Paired date-block uncertainty

All cities on a date can share weather conditions, market activity or information patterns. The uncertainty calculation first pairs model and baseline losses within each event, then gives equal weight to each calendar date:

$$\overline{\Delta} = \frac{1}{D}\sum_{d}\frac{1}{n_d}\sum_{e\in d}(L_e^{model} - L_e^{base}) \tag{17}$$

Equation 17. Date-paired score difference

Negative values favour the model. As Künsch (1989) develops in *The jackknife and the bootstrap for general stationary observations*, blocks can preserve local dependence when resampling dependent observations. Here a moving-block bootstrap resamples consecutive date means, concatenates sampled blocks and truncates to the original 18-date length. It uses 1,999 draws and percentile limits at 2.5 and 97.5 percent.

Three-day blocks are primary; one- and five-day blocks are sensitivity checks. With only 18 later dates, longer blocks leave few distinct stretches of history and seasonal change

may undermine stationarity. The resulting intervals should therefore be read as small-sample sensitivity evidence, not assured nominal coverage. Equal-date weighting also differs slightly from equal-event weighting because one later date is missing Chicago.

7 Empirical findings

7.1 Historical spreads and conditional basket gaps

Figure 2 summarises quoted spreads for the accepted noon panels. Each spread compares the bid and ask summaries at a completed hourly endpoint. It cannot reveal the size available at that spread or the order in which constituent quotes updated.

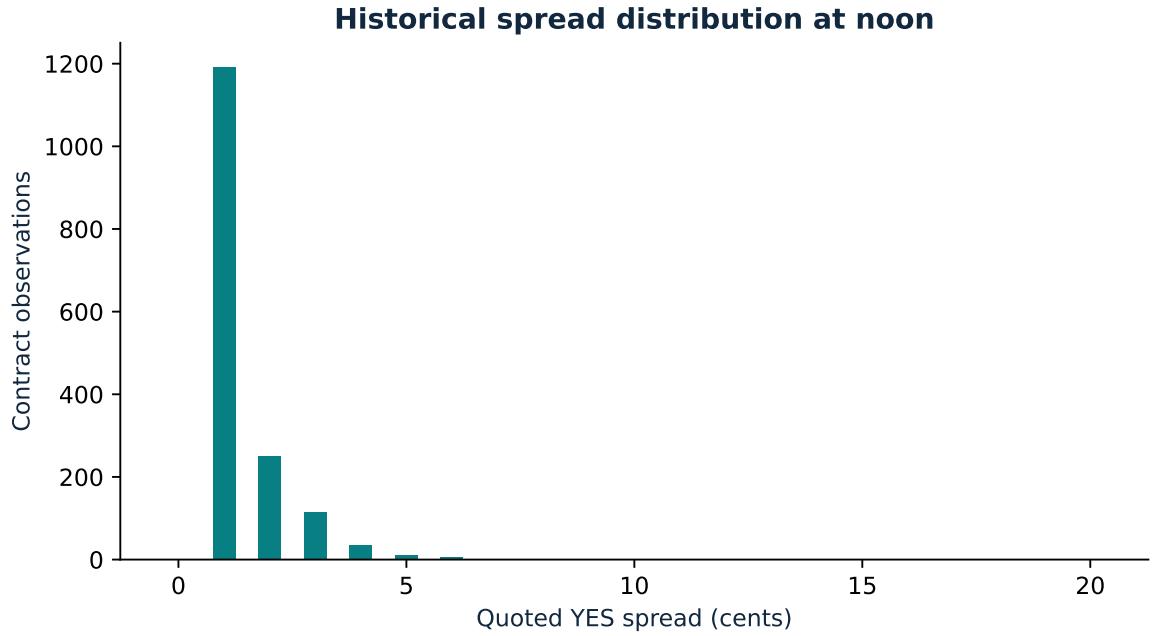


Figure 2. *Historical spreads from accepted noon events. Candle-close quotes have no accompanying depth or fill guarantee.*

The 268 accepted events generate 536 basket-side checks: one YES basket and one NO basket per event. Five YES baskets and 29 NO baskets have a positive gross ordinary-resolution margin. These 34 positive checks happen to occur in 34 distinct events, but the audit denominator is the number of side checks. Table 8 keeps the two sides separate.

Basket	Checks	Gross positive	Net positive	Max gross (cents)	Max net (cents)
YES	268	5	0	2.00	-2.13
NO	268	29	0	4.00	-0.11

Table 8. *Historical conditional basket discrepancies.*
One contract per leg; 268 events per side. Net values use the current direct-member rounding scenario. No historical fill or depth is asserted.

The largest gross margin is two cents for a YES basket and four cents for a NO basket. After the current direct-member fee scenario, no basket remains positive. The closest

NO result is still negative by 0.11 cents per unit basket, while the maximum net YES result is negative by 2.13 cents. Thus the fee adjustment is material even before adding uncertainty about contemporaneous liquidity and the applicable historical fee regime.

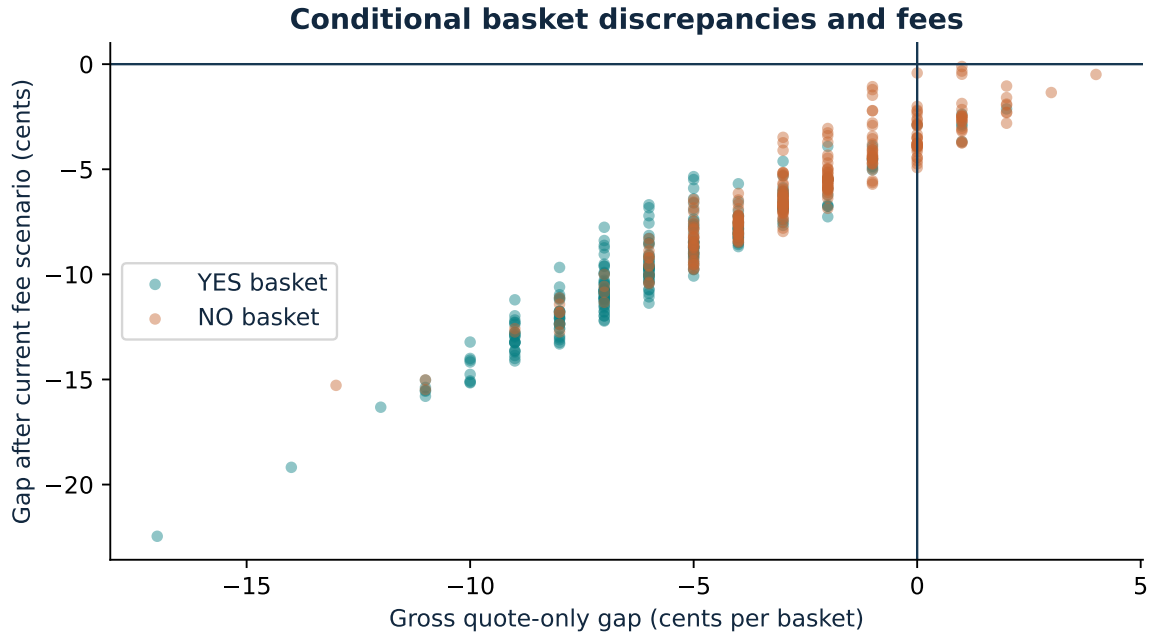


Figure 3. *Conditional integer-resolution basket gaps from hourly quotes; current fee scenarios do not establish historical execution or contractual arbitrage.*

Figure 3 illustrates why a gross discrepancy count cannot be presented as an arbitrage hit rate. It omits actual fills, depth and the admissible-state qualification unless those conditions are stated separately. The finding is that the retained quote sample contains small conditional discrepancies that do not survive the stated cost scenario. It is not a proof that every opportunity in these markets was absent.

7.2 Calibration fit and model rejection

The development fit gives $T = 1.034401$, close to the identity. Development logarithmic loss falls from 0.956481 to 0.956092, a small improvement. On validation, however, candidate loss is 0.819043 against 0.813803 for the baseline. The pre-established selection rule therefore retains $T = 1$.

Sample	Probability vector	Log loss (nats)	Brier score
Development	Baseline	0.956481	0.553895
Development	Candidate	0.956092	0.552713
Validation	Baseline	0.813803	0.464422
Validation	Candidate	0.819043	0.465765
Test	Baseline	0.718589	0.408273
Test	Candidate	0.722152	0.409355

Table 9. *Primary probability scores at noon.*
Candidate $T = 1.034401$; validation selected $T = 1$, so the selected model equals the baseline. Lower scores are better.

On the 53 later noon events, baseline log loss is 0.718589 and Brier score is 0.408273. The rejected candidate produces 0.722152 and 0.409355 respectively. Figure 4 displays both scores without interpreting the rejected candidate as the chosen forecast. The selected model's score difference from the baseline is exactly zero in mathematical terms; negligible floating-point differences are not evidence of performance improvement.

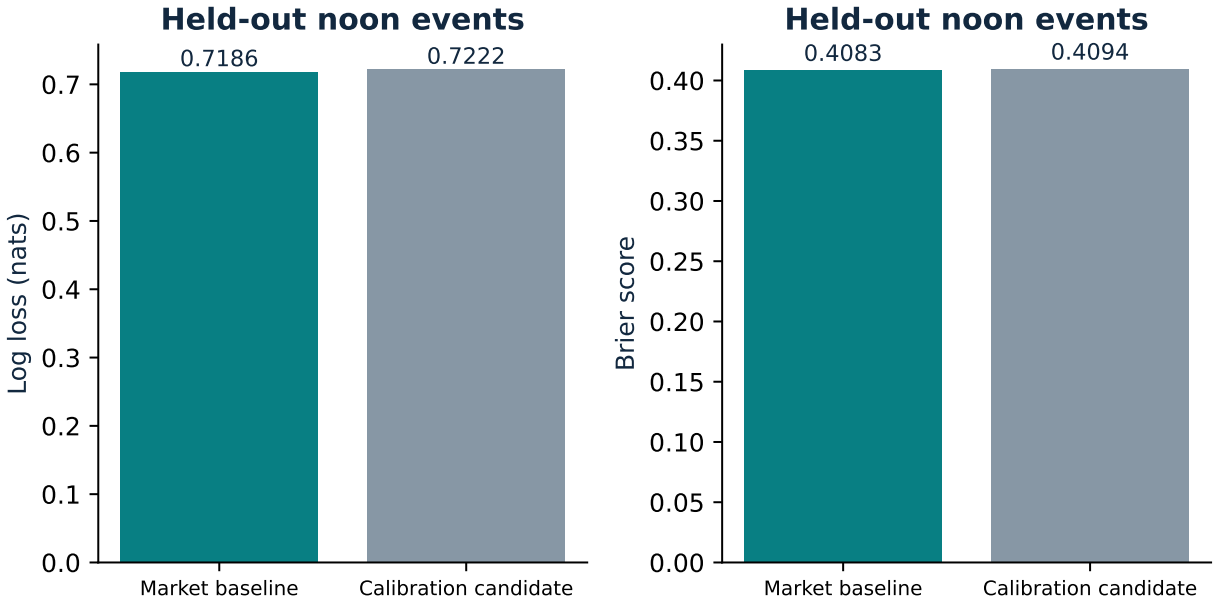


Figure 4. *Held-out scores for the market-derived baseline and the rejected calibration candidate. Lower is better; validation selected the baseline.*

The rejection is substantively useful. A more complex label does not imply a better forecast when the input is already the market's own probability-like vector. Temperature scaling can correct only a common concentration pattern. It cannot introduce meteorological information, resolve city-specific biases or repair the outcome definition. The observed failure limits this particular adjustment, rather than ruling out all probability models.

7.3 Reliability and uncertainty

The pooled reliability plot in Figure 5 groups later predicted probabilities and compares them with observed binary frequencies. It is descriptive because probabilities within an event sum to one and the contracts share a winner. Sparse high-probability bins and a short date span can make the curve unstable. The plot is not an independent-bin test of calibration.

The equal-date mean candidate-minus-baseline differences at noon are 0.003717 for log loss and 0.001198 for the Brier score. They differ slightly from subtracting the event-weighted scores in Table 9, because dates are equally weighted for inference. Table 10 reports the paired block intervals.

For log loss, the primary three-day interval runs from -0.000564 to 0.008773. The Brier interval runs from -0.000555 to 0.003494. Both include zero. One- and five-day alternatives also include zero at noon, as Figure 6 shows. These intervals do not rescue the rejected model: the selection decision was already made on validation, and lack of a statistically clear later difference is not evidence of useful expected trading profit.

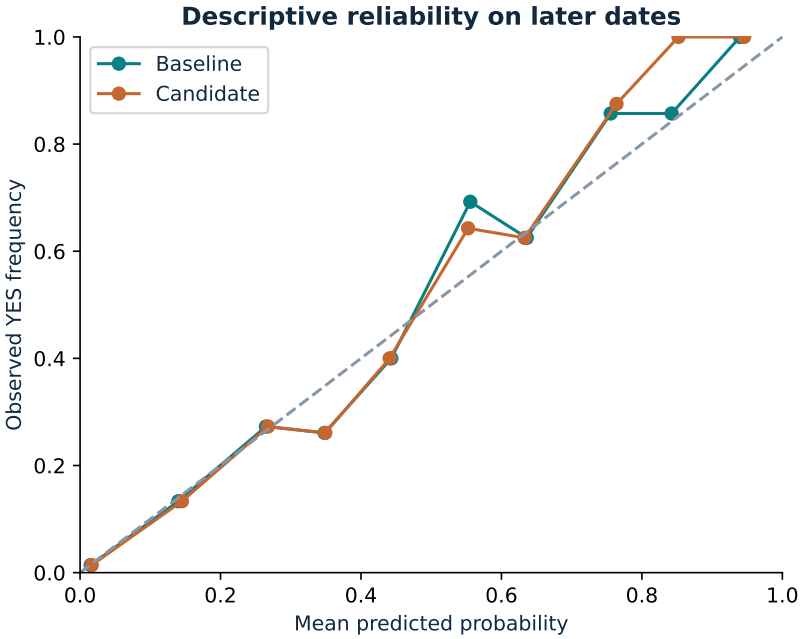


Figure 5. *Pooled reliability bins on held-out events. Contracts within an event are dependent; this curve is descriptive, not an independent-bin significance test.*

Metric	Block days	Mean difference	95% interval
Log loss	1	0.003717	$[-0.003279,\ 0.008874]$
Log loss	3	0.003717	$[-0.000564,\ 0.008773]$
Log loss	5	0.003717	$[-0.001122,\ 0.007645]$
Brier	1	0.001198	$[-0.001319,\ 0.003571]$
Brier	3	0.001198	$[-0.000555,\ 0.003494]$
Brier	5	0.001198	$[-0.000723,\ 0.002816]$

Table 10. *Paired date-block sensitivity for the rejected candidate.*
Candidate minus baseline, equal weight per date; 18 dates and 1,999 draws. Three days is primary. These are supplementary comparisons, not a revised selection rule.

7.4 Observation-time and city sensitivity

Later baseline log loss decreases from 0.879706 at 09:00 to 0.718589 at noon and 0.272615 at 15:00. The temperature outcome is more nearly realised later in the day, so the forecasting problem changes with the observation time. The accepted event sets also differ slightly. Table 11 therefore reports sensitivity rather than an estimated causal effect of waiting.

The noon-fitted candidate is transferred to these hours without refitting. It performs slightly worse on both later scores at all three hours. City-specific later results in Table 12 likewise favour the baseline in the reported averages. The small numbers of dates and cities prevent treating these comparisons as a broad survey of market calibration.

No later noon event passes the exploratory two-cent model-edge threshold. The diagnostic therefore contains zero hypothetical selections and no mean selected-trade payoff. A recorded total scenario payoff of zero reflects the empty set of selections; it is not a

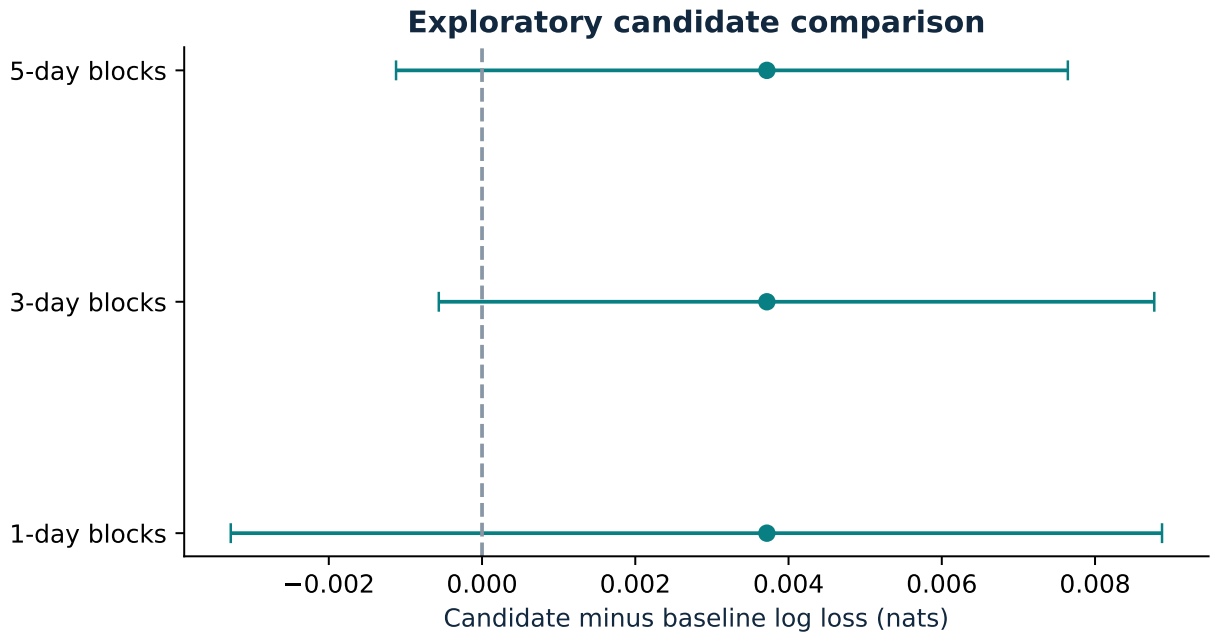


Figure 6. *Paired date-block confidence intervals for the rejected candidate on later dates; negative differences favour the candidate. The selected method is the baseline itself.*

Local time	N	Base log loss	Candidate log loss	Base Brier	Candidate Brier
09:00	54	0.879706	0.884391	0.480503	0.482405
12:00	53	0.718589	0.722152	0.408273	0.409355
15:00	53	0.272615	0.277643	0.162587	0.162864

Table 11. *Later-period sensitivity to observation time.*
The noon-fitted candidate is transferred unchanged. Different accepted panels and information horizons prevent a causal interpretation of cross-hour differences.

City	N	Base log loss	Candidate log loss	Base Brier	Candidate Brier
Chicago	17	0.847000	0.847490	0.439558	0.440986
Miami	18	0.669839	0.672819	0.375481	0.376419
New York	18	0.646061	0.653111	0.411518	0.412416

Table 12. *Later-period scores by city.*
All cities use the same fitted temperature and chronological split; no city-specific refitting.

zero-risk strategy or an achieved return. The probability analysis provides no evidence of profitable model-based trading in this sample.

7.5 Live liquidity and statewise screening

The live collection runs from 20:29:46.177 to 20:32:32.296 UTC on 17 October 2026. Twenty batch responses cover 36 contracts in six events. The responses have 20 distinct canonical payload hashes, showing that the retained batches are not identical copies. This does not establish how many order events occurred between requests.

Object	Count or estimate	Meaning
Batch responses / duration	20 / 166.12 s	One short collection session on 17 October 2026.
Individual books	720	36 contracts across 6 events, observed 20 times.
Two-sided / one-sided books	406 / 314	Spread summaries use the two-sided subset.
Median spread	1.00 cent	Displayed spread, not an executed transaction cost.
Median request span	5.307 s	HTTP collection span, not exchange latency.
Complete basket-side-size checks	378 of 720	Enough displayed depth; none has a positive net gap.
Checks rejected for depth	342 of 720	At least one leg cannot supply the required size.
Statewise portfolio checks	46	Complete two-sided event observations; no positive solution.

Table 13. *Recorded live-pilot audit.*
The 720 individual books and 720 basket checks are different denominators. Basket checks use two sides and three quantities per event per round.

Of the 720 individual market books, 406 are two-sided and 314 are one-sided. The median spread among two-sided books is one cent. The median HTTP request span is 5.307 seconds, exceeding the two-second observation screen, but this span measures collection uncertainty and includes network and service effects. It must not be described as exchange execution latency.

Figure 7 shows acquisition costs for the final observed contract whose midpoint is closest to 0.50. This deterministic descriptive selection avoids choosing the contract for a favourable future return. The cost curve rises as displayed levels are consumed. A fee-adjusted curve can be materially higher than the best ask even before unobserved execution slippage.

The 720 basket checks arise from six events, 20 rounds, two sides and three quantities: one, ten and one hundred contracts per leg. Of these checks, 378 have enough displayed depth and show no positive net gap; 342 fail because at least one leg lacks the requested quantity. These counts are distinct from the coincidentally equal number of individual book observations. Forty-six event-round observations have complete two-sided books

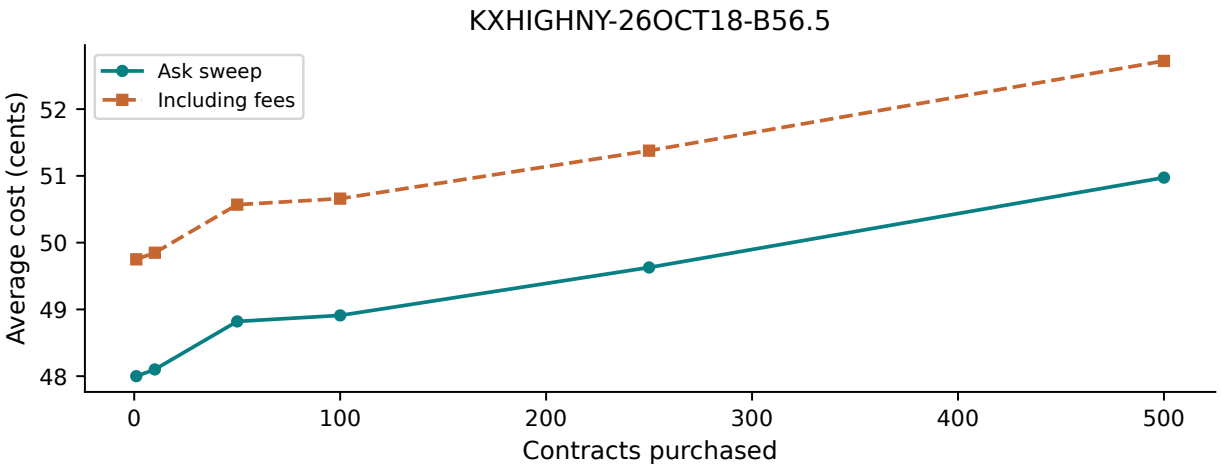


Figure 7. *Displayed-depth purchase costs for the final snapshot whose midpoint is nearest 0.50; the selection rule is descriptive and does not use future returns.*

and support the general statewise programme. None yields a positive relaxed or rounded-and-rechecked lowest-state payoff.

7.6 Exploratory imbalance diagnostic

There are 385 usable pairs of snapshot imbalance and next observed midpoint change. Their pooled correlation is 0.0355. Figure 8 retains the concentration of observations at unchanged midpoints and the variation across contracts rather than reducing the diagnostic to a fitted trading rule.

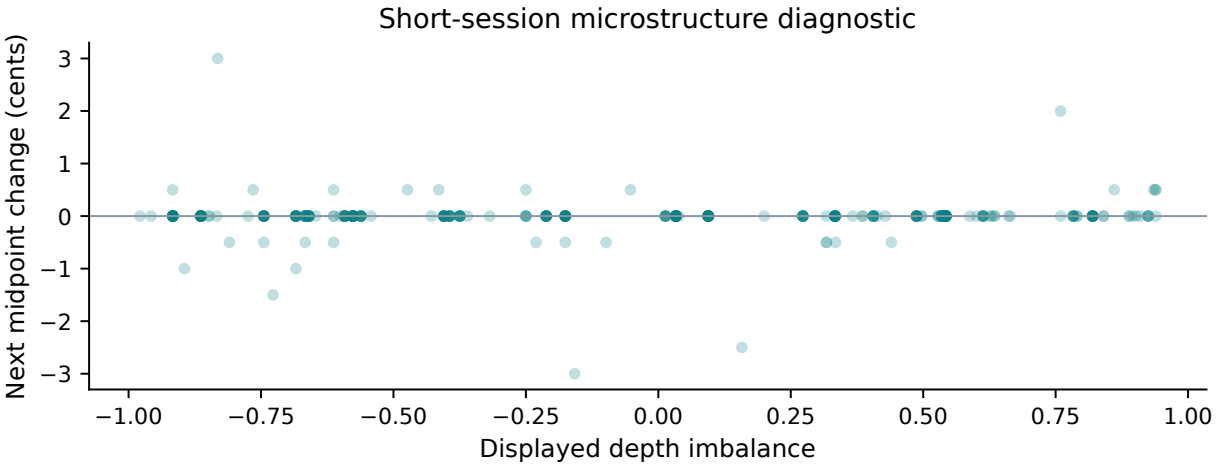


Figure 8. *Snapshot imbalance and the next observed midpoint change in the brief live session; these are not order-flow events or independent trading trials.*

These pairs repeatedly observe a small set of contracts over a few minutes. They are neither independent opportunities nor a representative time sample. No predictive model or significance claim is inferred from the correlation. A stronger study requires a longer prospective record with exchange-timestamped updates and explicit missing-data rules.

8 Synthetic validation and execution risk

8.1 Known-truth screening mechanisms

Synthetic validation isolates logical behaviour from the absence of empirical opportunities. Each of 500 probability draws is reused across six mechanisms. The probabilities are drawn from a six-dimensional Dirichlet distribution with concentration eight and shifted away from zero. Coherent asks add one cent to each component. A true supplied-state gap uses asks equal to 90 percent of the probability vector, while a deliberately small gross gap uses 99.5 percent and is removed by fees.

Other mechanisms hold the attractive basket arithmetic fixed but reduce depth below the requested ten contracts, impose a five-second observation span, or activate the exceptional-settlement block. These tests use a supplied exhaustive single-winner state set and known labels. They do not inherit the empirical temperature contract's unresolved state-domain qualification.

Mechanism	Checks	Correct	Eligible
Asynchronous observation	500	100.0%	0.0%
Coherent prices	500	100.0%	0.0%
Exceptional settlement	500	100.0%	0.0%
Fees remove the gap	500	100.0%	0.0%
Insufficient depth	500	100.0%	0.0%
True supplied-state gap	500	100.0%	100.0%

Table 14. *Known-truth synthetic screening checks.*
500 probability draws are paired across six mechanisms: 3,000 scenario evaluations, not 3,000 independent markets.

All 3,000 scenario evaluations produce the expected classification in this designed family. The count represents 500 paired probability draws across six conditions, not 3,000 independent markets. Perfect accuracy within constructed cases is evidence that these specified rules are implemented consistently. It is not a measure of sensitivity to every possible real contract error or an estimate of profitable trading frequency.

8.2 Incomplete multi-leg execution

A second experiment studies a six-leg basket whose asks are each 0.15 dollars. Its gross cost is 0.90 dollars and its raw quadratic fee is 0.05355 dollars. If all legs fill and exactly one pays one dollar, the terminal margin is 0.04645 dollars before any added entry-cost stress. The terminal winning leg is chosen uniformly, independently of the assumed fill process.

For a common per-leg fill probability ρ and K independent legs, the probability of a complete basket is

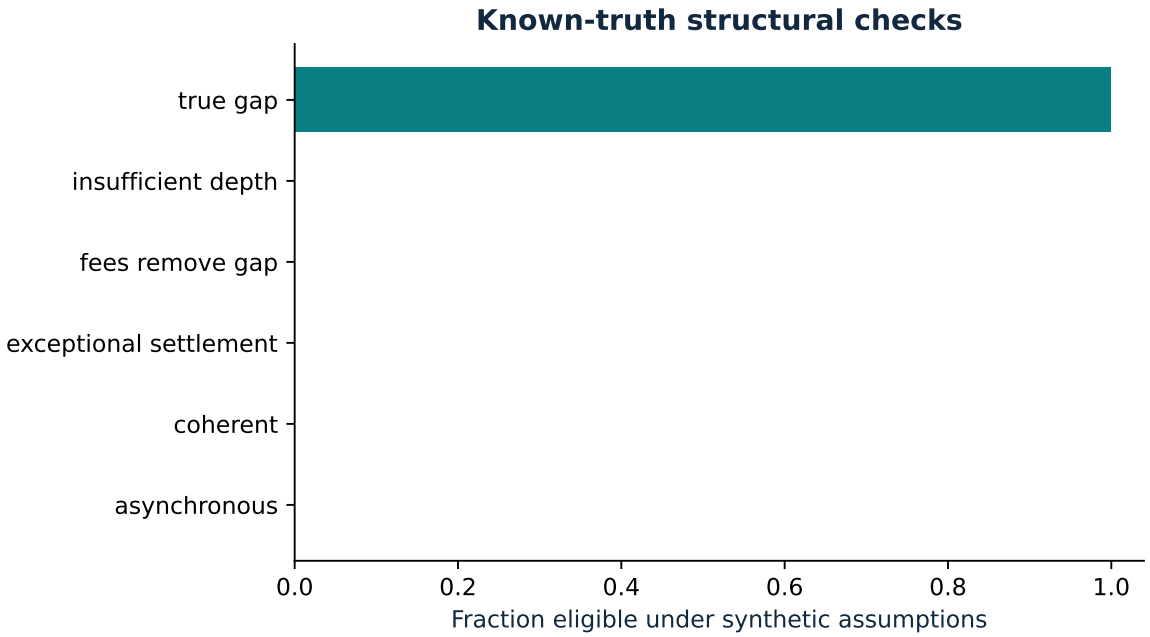


Figure 9. *Six synthetic mechanisms, 500 replications each. Passing known-truth tests verifies selected logic, not real-market profitability.*

$$\Pr(\text{all legs fill}) = \rho^K \tag{18}$$

Equation 18. Synthetic complete-basket fill probability

This is a synthetic independence assumption, not an empirical fill estimate. At $\rho = 0.95$ and $K = 6$, the analytical complete-fill probability is about 73.5 percent. A high individual-leg fill probability therefore need not imply near-certain basket completion.

The simulation uses 20,000 draws at each of five fill probabilities. The same draws are reused across four additional entry-cost settings, giving 20 scenario cells and 100,000 distinct fill draws. The added cost is charged per filled leg and is treated as an extra penalty; the raw fee remains calculated at the base ask. This is an explicit cost stress, not a reconstruction of repriced exchange orders.

Per-leg fill	All filled	Loss, base	Mean, base ($)	Loss, +1c	Mean, +1c ($)
0.50	1.5%	48.5%	0.0236	49.9%	-0.0064
0.70	11.7%	29.6%	0.0363	41.3%	-0.0057
0.85	37.2%	15.2%	0.0391	52.4%	-0.0118
0.95	73.1%	5.1%	0.0434	78.2%	-0.0135
1.00	100.0%	0.0%	0.0465	100.0%	-0.0135

Table 15. *Synthetic six-leg execution stress.*
20,000 draws per fill setting, paired across entry-cost scenarios. +1c adds $0.01 per filled leg while retaining the base raw fee. No empirical fill probability is estimated.

With a 95 percent per-leg fill probability and no extra cost, the simulated mean payoff is 0.0434 dollars, while 5.14 percent of trials lose money. Adding one cent per filled leg makes the mean negative and raises the simulated loss rate to 78.20 percent. At certain fill, the base basket earns the constructed 0.04645-dollar margin, but the one-cent penalty

makes every trial lose 0.01355 dollars. The negative result follows from six additional cents exceeding the original margin.

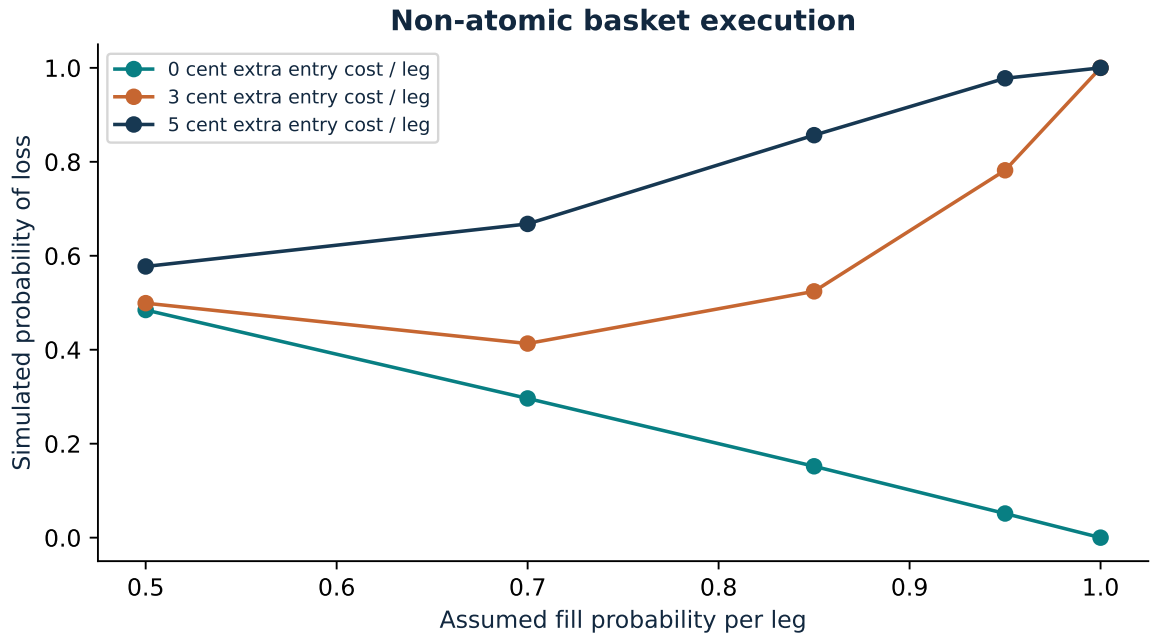


Figure 10. *Synthetic independent leg-fill stresses. A positive expected payoff can coexist with losses when an incomplete basket remains exposed.*

Figure 10 illustrates that execution risk and expected value answer different questions. Incomplete baskets can lose even when their average payoff is positive. The experiment excludes correlated fills, queue priority, dynamic hedging, cancellation races, inventory netting, funding costs and settlement exceptions. It demonstrates a mechanism rather than forecasting a realised trading distribution.

9 Discussion, research reflections and limitations

9.1 What the combined evidence establishes

The historical sample reveals small conditional quote gaps that fail the current-fee scenario. The live pilot shows how missing sides, depth and observation timing constrain a scanner. The synthetic cases verify selected logic under deliberately sufficient payoff assumptions. These components answer different questions and cannot be pooled into a profitability estimate.

No surviving candidate is evidence about this sample, not proof of market efficiency. Other contracts, faster observations or different costs could change the result. A future positive screen would still require complete state coverage, adequate size and execution evidence.

The calibration failure supports retaining the baseline, not a claim of perfect market calibration. External weather forecasts or richer models require a new development and validation exercise; the present later scores cannot become a model-selection dataset without changing their evidential status.

9.2 Acquisition failures and implemented corrections

The initial batch-candle request used 80 tickers across eight days and failed against the endpoint's 10,000-candle potential-grid cap. The limit concerns possible ticker-time combinations, not just the number of nonempty observations. Reducing batches to 32 tickers gives at most 6,144 hourly combinations over eight days. This repaired collection while preserving the planned event window.

Archived and current candle responses also used different field names for decimal-dollar values. A magnitude-based guess could confuse a dollar value with cents. The adapter instead recognises the two documented schemas explicitly and requires a string-valued dollar field. Missing archived records are routed through the historical endpoint rather than treated as evidence that the market never existed. These changes address data representation and retention, not the model's fitted outcome relationships.

A concurrent append left one source snapshot absent from the original manifest. The raw snapshot itself remained intact. Rebuilding the index by enumerating stored source files recovered a complete 521-file register, with snapshot and canonical-payload hashes. This correction repairs the current audit trail; it does not claim that the original parallel append mechanism was made transactional. A future collector should serialize manifest writes or use an atomic indexed store.

9.3 Contract and model assumptions that failed

The initial intuition that six integer-labelled ranges automatically establish a guaranteed one-dollar YES basket did not survive the contract audit. The full-precision comparison model permits an additional all-zero YES row, and the exceptional settlement clause remains outside the simple binary partition. The mitigation is to preserve integer-resolution basket arithmetic as conditional, expand the ordinary state matrix and withhold an unconditional certificate. The remaining limitation is uncertainty about the exact admissible domain of each source report and the exceptional payout set.

The fitted calibration candidate also failed its validation criterion. It marginally improved development loss but worsened validation loss. The implemented mitigation was rejection, not an expanded search over models on the later sample. The final selected forecast remains the baseline. Retaining the rejected candidate in explicitly supplementary tables makes the failure inspectable without changing the selection rule.

One-sided books exposed a presentation issue during browser review: an absent side could produce an infinite displayed spread if an empty array was reduced numerically. The explorer now labels the spread unavailable and defaults to a two-sided contract. A separate plotting-library title change initially omitted scientific axis labels; explicit title objects and adjusted camera distance restored legibility. These presentation fixes did not alter the frozen empirical predictions or numerical analyses.

9.4 Verification and residual risk

Table 16 records the completed checks. The analytical tests cover complementary quotes, decimal costs, fee rounding, insufficient liquidity, omitted states, optimisation, calibration

and dependence-aware scoring helpers. They exercise economically meaningful failure cases rather than only checking whether scripts return an output.

Layer	Check	Recorded result
Analytical tests	Books, costs, payoff states, probabilities and bootstrap helpers	45 tests passed
Synthetic mechanisms	Six designed scenarios, paired probability draws	3,000 checks passed
Notebook execution	Eight nbformat-4 notebooks	31 code cells executed
Source provenance	Stored snapshot hashes verified during offline replay	521 snapshots verified
Browser behaviour	Interactive controls, mathematical rendering, downloads and mobile width	24 checks passed
Analytical freeze	Inputs, implementation and frozen probabilities	Unchanged during paper assembly

Table 16. *Research and presentation verification.*

Verification establishes reproducibility and selected implementation properties; it does not certify market profitability or complete settlement-state coverage.

Verification does not establish permanent endpoint compatibility, complete settlement coverage or actual fills. The scanner supports the audited series and rejects unsupported fee metadata. Historical book reconstruction, queue estimation, order submission and management after partial execution remain outside its scope.

The settlement filter further limits external validity. Excluding other resolution types conditions scoring on a single-winner universe. A broader project should model those contingencies or declare its admissible population before outcomes are observed. Normalised probability scores cannot certify exceptional settlement behaviour.

9.5 Data access and publication status

The Kalshi (2026b) developer agreement distinguishes access from permitted use. Its provisions restrict API-data collection and storage to facilitating a member's own trading, restrict sharing without written authorisation, and require approval for specified public disclosures. The ability to retrieve an unauthenticated response is therefore not treated as an open-data licence or as proof that research retention and publication are authorised.

The retained cache and empirical outputs are private working material, with permission for their intended retention and public release unresolved. This paper does not represent written authorisation as obtained. An eventual public package requires a documented rights decision covering the paper, empirical aggregates, graph payloads and source data. Original code and synthetic demonstrations can be prepared separately, but that separation does not by itself clear every proposed public use.

9.6 A stronger subsequent research design

A prospective extension should freeze the scanner and inclusion rules before collecting a longer permitted sample. Exchange-timestamped updates would support event-based

microstructure analysis; execution records or a validated queue model would be needed for fill estimates. REST response spans cannot supply that evidence.

External forecasts would require verified issue times, location definitions and settlement-source compatibility. A forecast published after a decision cannot enter its earlier information set. Model complexity should be chosen on new validation data, with later observations reserved for final evaluation. These changes address the limits exposed by this study.

10 Conclusion

This study builds a reproducible prediction-market research engine around three connected questions: what contracts pay, what positions cost to acquire, and how well estimated probabilities transfer to later outcomes. It makes the distinctions operational through explicit payoff states, depth sweeps, fee calculations, chronological model selection and synthetic execution tests.

In 536 historical conditional basket-side checks, 34 gross discrepancies are positive and none survives the stated current-fee scenario. The short live pilot produces no positive cost-adjusted basket or statewise portfolio candidate. The calibration parameter fitted on development data fails validation, so the final forecast retains the market-derived baseline. These findings provide no evidence of realised trading profitability and do not imply that every prediction market is efficient.

The practical contribution is a disciplined method for rejecting unsupported opportunities and preserving the reasons for rejection. Missing depth, uncertain observation timing, incomplete payoff coverage, weak probability improvements and partial execution each require different remedies. Keeping those mechanisms visible produces an auditable foundation for future market-microstructure and probability research.

References

Avellaneda, M. and Stoikov, S. (2008) 'High-frequency trading in a limit order book', Quantitative Finance, 8(3), pp. 217–224. doi: 10.1080/14697680701381228.

Boyd, S. and Vandenberghe, L. (2004) Convex optimization. Cambridge: Cambridge University Press. doi: 10.1017/CBO9780511804441.

Brier, G.W. (1950) 'Verification of forecasts expressed in terms of probability', Monthly Weather Review, 78(1), pp. 1–3. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.

Cont, R., Kukanov, A. and Stoikov, S. (2014) 'The price impact of order book events', Journal of Financial Econometrics, 12(1), pp. 47–88. doi: 10.1093/jjfinec/nbt003.

Glosten, L.R. and Milgrom, P.R. (1985) 'Bid, ask and transaction prices in a specialist market with heterogeneously informed traders', Journal of Financial Economics, 14(1), pp. 71–100. doi: 10.1016/0304-405X(85)90044-3.

Gneiting, T. and Raftery, A.E. (2007) 'Strictly proper scoring rules, prediction, and estimation', Journal of the American Statistical Association, 102(477), pp. 359–378. doi: 10.1198/016214506000001437.

Guo, C., Pleiss, G., Sun, Y. and Weinberger, K.Q. (2017) 'On calibration of modern neural networks', Proceedings of the 34th International Conference on Machine Learning, 70, pp. 1321–1330. Available at: https://proceedings.mlr.press/v70/guo17a.html (Accessed: 17 October 2026).

Hasbrouck, J. (2007) Empirical market microstructure: The institutions, economics, and econometrics of securities trading. New York: Oxford University Press. doi: 10.1093/oso/9780195301649.001.0001.

Kalshi (2026a) API documentation. Available at: https://docs.kalshi.com/ (Accessed: 17 October 2026).

Kalshi (2026b) Developer agreement. Version 1.1. Available at: https://assets.kalshi.com/Kalshi-Developer-Agreement.pdf (Accessed: 17 October 2026).

Kalshi (2026c) Fee rounding. Available at: https://docs.kalshi.com/getting_started/fee_rounding (Accessed: 17 October 2026).

Kalshi (2026d) Fee schedule. 7 July. Available at: https://kalshi.com/docs/kalshi-fee-schedule.pdf (Accessed: 17 October 2026).

Kalshi (2026e) GLOBALTEMPERATURE: Contract terms and conditions. Available at: https://assets.kalshi.com/contract_terms/GLOBALTEMPERATURE.pdf (Accessed: 17 October 2026).

Künsch, H.R. (1989) 'The jackknife and the bootstrap for general stationary observations', The Annals of Statistics, 17(3), pp. 1217–1241. doi: 10.1214/aos/1176347265.

Wolfers, J. and Zitzewitz, E. (2006) Interpreting prediction market prices as probabilities. NBER Working Paper No. 12200. Cambridge, MA: National Bureau of Economic Research. doi: 10.3386/w12200.

A Reproducibility and implementation

A.1 Notebook sequence and recorded inputs

Table 17 maps the executed notebooks to the research stages. The notebooks contain 31 executed code cells in total, with mathematical exposition, tables and figures retained alongside their calculations. The HTML research review provides an accessible reading route and downloads of the individual notebook files.

Notebook	Research stage	Code cells
00	Questions, contracts and research design	3
01	Public API data and provenance	4
02	Order books, fees and liquidity	4
03	Payoff constraints and arbitrage	4
04	Probability modelling and model selection	4
05	Chronological validation and model-based value	4
06	Execution risk and research reflections	4
07	Synthesis and publication registers	4

Table 17. *Executed notebook sequence.*
Each notebook retains executed outputs. The HTML review groups them by milestone and provides individual downloads.

The project requires Python 3.12 or later and the pinned NumPy, pandas, SciPy and Matplotlib dependencies in `requirements.txt`. From the project root, `python scripts/reproduce.py` verifies retained hashes, executes the analytical tests and recreates the derived outputs and notebooks from the saved sample. It issues no new network requests. The optional `--web` flag also rebuilds the offline HTML pages using the recorded Node.js dependencies.

The scanner defaults to the final recorded observation. Run `python` on scripts/scan.py with the options `--series KXHIGHNY` and `--quantity 10` for a reproducible scan of ten contracts per basket leg. A separate live option exists for a new permitted public observation; it is not invoked during paper compilation. The current historical collector is configured for the bounded 90-date research window, not an unrestricted future backtest.

The model freeze was recorded at 20:41:16.189 UTC on 17 October 2026, before the later score calculation. It identifies the development fit, validation decision, feature hash, prediction hash and analytical implementation hashes. The prediction file's SHA-256 digest is

ce2b9bfb620ff5bf12d027f13d72f1f54831284b8443cf6d8b2a5c9b83bcb83f

The paper builder checks the analytical freeze and reads existing outputs. It does not reacquire data, refit the temperature parameter, change the selection rule or replace the retained predictions. The paper source package contains the typesetting files and vector exhibits needed to compile the document independently of model estimation.

A.2 Fee calculation excerpt

Code listing 1 is extracted from the implemented fee routine. It assumes the caller has supplied an ordered sequence of fills and a fee policy. Decimal helpers and policy definitions remain in the accompanying Python module. The recurrence preserves the distinction between the trade fee, balance rounding and an order-level rebate accumulator.

```
def execution_fee(fills,policy=FeePolicy()):
    """One order, enumerated fills, per-fill six-decimal fee and rounding accumulator.

    The caller specifies fill granularity. A price level is not necessarily one fill.
    Return is a scenario, never a reconstruction of undisclosed counterparties.
    """
    if policy.coefficient<0 or policy.multiplier<0 or policy.balance_precision<=0:raise
     ↪  ValueError('Invalid fee policy')
    accumulator=ZERO; total=ZERO
    for p,q in fills:
        p,q=decimal(p),decimal(q)
        if not ZERO<=p<=ONE or q<ZERO:raise ValueError('Invalid fill')
        raw=policy.coefficient*policy.multiplier*q*p*(ONE-p)
        trade=ceil_grid(raw,D('0.000001'))
        revenue=-p*q
        aligned=floor_grid(revenue-trade,policy.balance_precision)
        rounding=(revenue-trade)-aligned
        accumulator+=rounding
        eligible=floor_grid(accumulator,policy.balance_precision)
        cap=floor_grid(trade+rounding,policy.balance_precision)
        rebate=min(eligible,cap)
        total+=trade+rounding-rebate;accumulator-=rebate
    return total
```

Code listing 1. Fill rounding and rebate accumulator

The assumed fill sequence remains an input to this calculation. Aggregated public depth does not disclose how many counterparties would fill an order. The implementation can compare fill-granularity scenarios but cannot infer the hidden matching sequence. This is why a deterministic fee routine does not by itself make the observed book an executable backtest.

A.3 Statewise optimisation excerpt

Code listing 2 implements the finite-state linear programme. Rows of the supplied matrix represent coherent underlying states, and columns represent positions. The objective is the lowest statewise net payoff under a cost budget and displayed quantity bounds. A numerical check verifies the solver's claimed minimum against the resulting state payoffs.

```
def statewise_portfolio(payoffs,costs,depth,budget=100):
    """Continuous long-only screening LP. Fees must be included in unit costs.

    Solves max g s.t. A q - c'q >= g in each supplied state. It is a model
    screen, not a rounded, funded or synchronously executable order basket.
    """
    A=np.asarray(payoffs,float);c=np.asarray(costs,float);d=np.asarray(depth,float)
    if A.ndim!=2 or A.shape[1]!=len(c) or c.shape!=d.shape or not np.all(np.isfinite(A))
     ↪  or not np.all(np.isfinite(c)) or not np.all(np.isfinite(d)) or np.any(c<0) or
     ↪  np.any(d<0) or budget<0:
        raise ValueError('Invalid payoff problem')
    B=np.column_stack((c[None,:]-A,np.ones(A.shape[0])))
    B=np.vstack((B,np.r_[c,0]));rhs=np.r_[np.zeros(A.shape[0]),budget]
    fit=linprog(np.r_[np.zeros(len(c)),-1],A_ub=B,b_ub=rhs,
                bounds=[(0,x) for x in d]+[(0,None)],method='highs')
    if not fit.success:raise RuntimeError(fit.message)
    q=fit.x[:-1]; profits=A@q-c@q
    if profits.min() < fit.x[-1]-1e-7:raise AssertionError('LP payoff validation failed')
    return dict(quantities=q.tolist(),cost=float(c@q),worst_profit=float(profits.min()),
     ↪  state_profits=profits.tolist())
```

Code listing 2. Statewise portfolio screening

The live analysis subsequently floors quantities and recomputes costs using the depth and fee routines. That recheck is separate from this continuous optimisation function. Settlement contingencies and non-simultaneous observations are also evaluated outside the numerical solver; they cannot be inferred from an optimisation success flag.

A.4 Chronological calibration excerpt

Code listing 3 fits the single temperature parameter using only the supplied development events. Validation selection is performed separately, and the resulting prediction file is frozen before later scores are calculated.

```
def fit_temperature(events):
    if not events:raise ValueError('No development events')
    loss=lambda
     ↪  logt:np.mean([scores(calibrate(e['p'],np.exp(logt)),e['winner'])['log_loss'] for
     ↪  e in events])
    result=minimize_scalar(loss,bounds=(np.log(.25),np.log(4)),method='bounded',options=
     ↪  {'xatol':1e-10})
    if not result.success:raise RuntimeError('Calibration did not converge')
    return
     ↪  {'temperature':float(np.exp(result.x)),'development_log_loss':float(result.fun)}
```

Code listing 3. Development-only temperature fitting

The log-temperature parameterisation ensures positivity within the fixed bounds. The objective is the mean logarithmic score, with no later-period outcome passed to the fitting function. The wider pipeline remains retrospective because the source history was acquired after the events; function-level separation is a reproducibility measure rather than proof of prospective blinding.

B Interactive research explorers

B.1 Recorded liquidity in three dimensions

The offline `interactive-explorers.html` file provides three views and CSV exports. Its liquidity view plots ask price, elapsed collection time and cumulative quantity, alongside two-dimensional acquisition costs. Figure 11 uses the same saved observations. The default contract is selected by its final midpoint's distance from 0.50. Lines join price levels within snapshots, not trades or unobserved orders between snapshots.

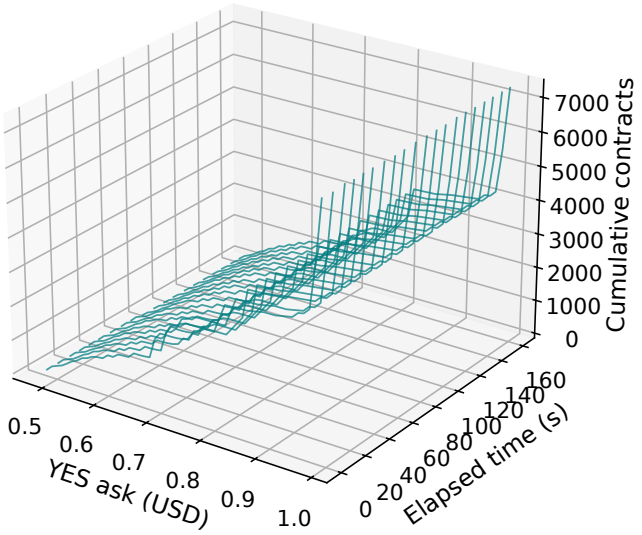


Figure 11. *Recorded order-book depth for KXHIGHNY-26OCT18-B56.5. Each line is one REST observation; the surface between observations is not reconstructed.*

B.2 Calibration sensitivity

The probability explorer varies temperature for a historical event, with axes for outcome rank, temperature and transformed probability. A companion chart compares the baseline with the slider setting and marks the subsequent winner. Figure 12 uses the first later-period New York event. The slider explores sensitivity; it neither refits a model nor revises the selected $T = 1$.

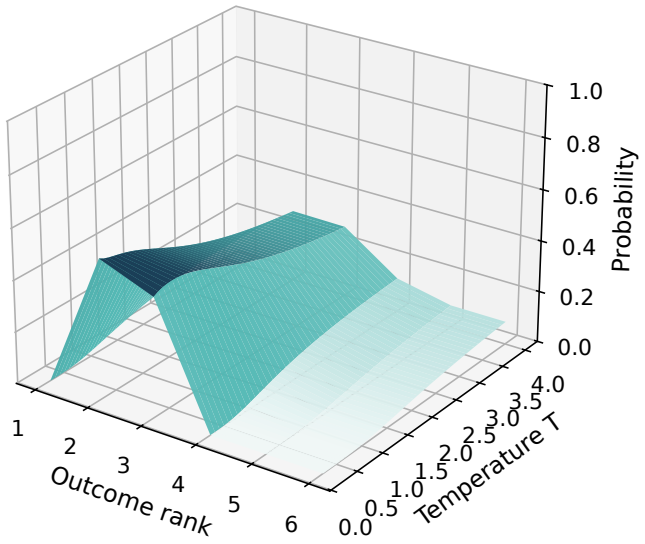


Figure 12. *Calibration sensitivity for KXHIGHNY-26SEP29, the first later-period New York event. T is varied for illustration; the selected research model remains T = 1.*

B.3 Synthetic basket execution

The synthetic laboratory varies basket discount, legs and fill probability under an exhaustive single-winner partition, independent fills and raw fees. Figure 13 sums binomial fill probabilities and the chance that the winning leg is absent. Its threshold reflects fees consuming the margin. The companion chart separates expected payoff from loss probability. Settlement exceptions, correlated fills, funding and dynamic hedging are excluded.

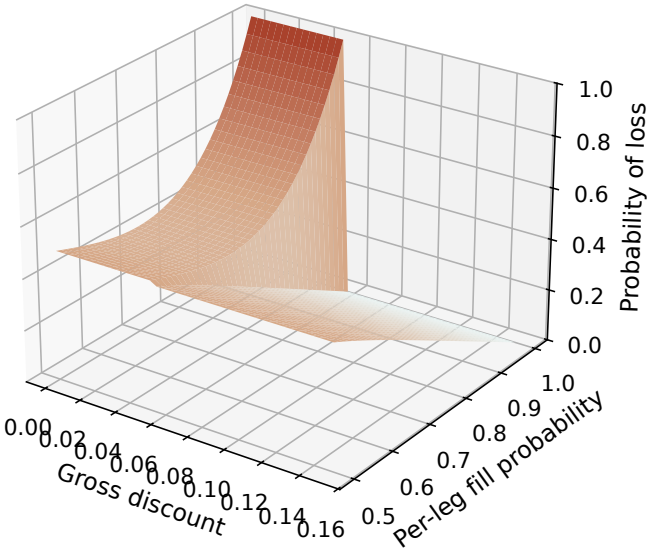


Figure 13. *Synthetic six-leg loss probability under independent fills, equally likely winning outcomes and raw quadratic fees. This analytical surface is not an empirical return forecast.*